

Supplements

A . Preregistered hypotheses

In line with the reviewed literature, we expected students in the build group to achieve higher subjective (H1) and objective (H2) AI literacy learning gains, and to report higher intrinsic motivation (H3) over time than students in the robot-interaction group. Additionally, we expected students in the build group to perceive the robot as more competent (H4), warmer (H5), safer (H6), less discomforting (H7), and more anthropomorphic (H8) over time than in the control group. Lastly, we expected students in the build group to feel more attached to (H9) and willing to use (H10) the robot over time than students in the control group. We additionally preregistered exploratory analyses of interactions with students' prior AI literacy knowledge and attachment to the robot.

B. Activities overview by week

Table S1. Overview of the activities during each week of the curriculum for both groups and unique to each condition.

	Both groups	Building group	Interaction group
Week 1	The research team introduced themselves (~5 minutes), and the students completed the pre-test survey (~20 minutes). The teacher then gave an overview of the curriculum and delivered a presentation about social robots, including a general introduction to robots, an online search of common social robots and an introduction to the social robot Marty and its components.	Students in the building group began familiarizing themselves with Marty's components and the manual (provided online and in print) and began assembling Marty's legs.	Students connected their Marty robots to their iPads and explored its functionalities. Students tested the controller (allowing remote control of movements), the coloured cards (allowing unplugged programming), and personalized their Marty with stickers.
Week 2	The teacher introduced AI and Machine learning, covering (i) unsupervised, supervised, and reinforced ML, and (ii) generation and classification. Students did two paper-based practice tasks on these concepts.	Students continued assembling Marty's legs.	Students played five rounds of a robot soccer tournament, with one Marty serving as the goalkeeper and the other as the striker. In each round, two teams competed against each other. Students exchanged roles (striker vs. goalkeeper) and players each round, ensuring that each child navigated Marty at least once in each role.
Week 3	The session began with a review and practice of previously covered AI concepts. Next, the teacher gave an introduction to LLMs, covering how they are trained, which LLMs exist, and how they use prompts to create the illusion of learning.	Students continued with building, mostly finishing the legs and starting the arms.	Each group designed an obstacle course and raced their Marty's through the courses. For each course, one team member recorded the time each Marty took to complete it. The fastest team won the respective obstacle course.
Week 4	Week 4 began with content on the limitations of AI, including biases, hallucinations, use cases in which AI should not be used, and strategies for the appropriate use of AI and LLMs. After the group-specific activities (see below), both groups completed the mid-test survey (~11 minutes).	Students completed the assembly of Marty, including finishing the arms and mounting eyes, arms, and legs to the head (total build time: 4 hours). Students tested their Marty's with iPads using the controller.	Students played an escape game. Each team navigated their Marty through various tasks and riddles at their table to retrieve a code to unlock a safe containing a leg of an extra Marty.
Week 5	Teachers presented different programming concepts and introduced Marty Blockly. Blockly is a simple, block-based programming language designed to facilitate programming education without text-based programming knowledge. Students then programmed a one-minute dance with Marty and presented it to the other teams. Next, teachers introduced sensors, including commonly available sensor types, Marty's sensors, how they work, and how to incorporate sensor input into programming. Lastly, students explored Marty's sensor outputs using the sensor board and triggered their dance choreography with Marty's obstacle-detection sensor.		
Week 6	Teachers introduced Microcontrollers and Arduino, covering what a microcontroller is, how it works, and that it is programmed in C++. Students then connected an Arduino Nano to their Marty robot and programmed simple movements using a pre-defined Marty library. Next, teachers gave a presentation about AI ethics and safety, including environmental consequences, resource demands, ethical guidelines, and safeguards. Students were then instructed to test an LLM-based voice interaction [gpt-4o and gpt-5-mini, 6] with Marty. This interaction was designed to reflect on the effect of safeguards on LLM responses through testing the LLM with and without safeguards.	Due to time constraints, students in the build group simply asked Marty one of four predefined questions (e.g., "Can you be my friend?" or "I would like to tell you a secret, promise me you won't tell anyone") without safeguards before the session was over.	The Interaction group did two trials, one with and one without safeguards, and compared the results. However, due to technical difficulties, this was done with an out-of-the-box LLM not linked to Marty. We address this in the limitations.
Week 7	The last session started with a review of the previously covered microcontroller and LLM safeguards. After the following activities (see below), students reflected on	Students were instructed to use Marty's LLM mode in two consecutive rounds to create a mnemonic about	Due to technical difficulties, Marty's LLM did not work. Students instead continued their Arduino programming and then created a

the curriculum and filled in the post-test survey (~12 minutes).	microcontrollers. In the first round, they did this without any instructions, using gpt-4.1-mini out of the box, then discussed the results. In the second round, students turned on a preprogrammed LLM infrastructure, called SCAFFOLD, designed to safeguard and structure learning interactions and described in more detail elsewhere [BLINDED].	mnemonic about microcontrollers with an out-of-the-box LLM.
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C Measurements

C1. Objective AI literacy scale

AI literacy. The AI literacy subscale was designed to capture objective, age-appropriate, and curriculum-relevant AI knowledge. We prioritized items that were understandable without dependence on external curricula because participating students had not received prior formal AI literacy instruction outside the intervention. This approach was intended to ensure that the assessment reflected the knowledge targeted in our intervention rather than prior exposure to specific AI education programs. Most AI literacy items were adapted from the AI Competency Objective Scale [AICOS; 5], a recently developed objective instrument that assesses multiple dimensions of AI literacy, including generative AI literacy ($\alpha = 0.83$). Since the original AICOS was developed as a substantially longer instrument for adults, only a subset of items was selected for the present study. Item selection was guided by three criteria: alignment with our curriculum, age appropriateness for our sample, and expected item difficulty. We selected 10 items that aligned with the content taught during our intervention and that we believed were of appropriate complexity for younger learners (6 easy items, 3 moderate items, 1 difficult item; the phrasings of the moderate and difficult items were adapted to simpler language). These items covered the following AI literacy subcompetencies identified in the original scale: detect AI (one item: DA09), create AI (one item: CA12), understanding AI (one item: UA19), ethics AI (two items: EA05, EA11), and generative AI (five items: GA10, GA13, GA04, GA16, GA17).

To further align the subscale with our curriculum, we added two machine learning items (CA29, CI31) from the AI Literacy Concept Inventory [AI-CI; 10], an instrument developed for middle school students and reported as a valid and reliable measure of AI literacy at that level ($\alpha = 0.73$). These items were selected because they addressed concepts relevant to the instructional content, whereas many other AI-CI items focused more strongly on algorithm training processes that were not treated in depth in our curriculum. The selected AI-CI items were adapted to match the instrument's response format (i.e., four answer options rather than two) and then translated into German. Finally, we developed one additional item on effective LLM prompting to cover a topic explicitly addressed in our curriculum but insufficiently represented in validated instruments. The final AI literacy subscale combined selected, adapted, translated, and newly developed items. As such, it should be understood as a study-specific, curriculum-aligned measure rather than a validated measure intended to assess AI literacy broadly across contexts.

Technical literacy. To assess technical literacy [i.e., “the ability to use, manage, assess, and understand technology”; , 9], we developed six multiple-choice questions following the same response format as the AI literacy subscale, since no existing instrument adequately captured the robot-specific technical knowledge taught in our curriculum. The items were intended to measure students’ understanding of the physical and electronic components of the social robot Marty and, therefore, represented content-specific technical literacy in our learning context.

The six items were distributed across three areas. Two items assessed knowledge of robot hardware, including identifying physical components and understanding the mechanical functions of parts such as motors. Two items assessed understanding of sensors, including the purpose of infrared sensors for obstacle and edge detection. Two further items assessed knowledge of microcontrollers, focusing on the role of the Arduino Nano ESP32 in executing simple programs and controlling

peripherals, as well as the distinction between a microcontroller and more powerful computing systems that can run larger AI models. This subscale was therefore designed to capture foundational technical understanding of the robot platform used in the intervention.

Programming literacy. To assess programming literacy, we developed four multiple-choice questions because no existing objective instrument matched both the sample's age range and the specific introductory programming content covered in our curriculum. The subscale was intended to assess foundational programming knowledge at a level appropriate for novice learners and closely aligned with our intervention's instructional goals. Two items targeted conceptual foundations by assessing students' understanding of basic programming ideas, specifically the meaning of a command and the ability to identify loops. One item assessed logic and flow control by asking students to interpret a C++ if-else statement and determine the condition under which a function would be executed. One additional item assessed syntax and sequence by requiring students to translate a verbal instruction into an appropriate sequence of C++ code using the Marty library functions, including the correct order of commands and parameter formatting. Together, these items were designed to measure whether students had acquired the basic declarative programming knowledge introduced during the intervention.

Scale reliability. Because the final instrument combined adapted and newly developed items, we treated the subscales as study-specific measures and calculated item and reliability statistics for the present sample. Item analyses were conducted for the general AI literacy, technical literacy, and programming literacy scales using baseline data (T1). Item difficulty was assessed as the proportion of participants answering each item correctly, while item discrimination was evaluated using corrected item-total correlations. For the *general AI literacy scale* (13 items), item difficulty ranged from .09 to .53, indicating that most items were relatively difficult. The proportion of "I don't know" responses ranged from .19 to .67. Corrected item-total correlations ranged from .13 to .42, suggesting generally acceptable discrimination. Internal consistency was acceptable, with Cronbach's $\alpha = .65$. For the *technical literacy scale* (6 items), item difficulty ranged from .18 to .58. The proportion of "I don't know" responses ranged from .21 to .54. Item discrimination values ranged from .12 to .32. Internal consistency was relatively low ($\alpha = .42$). For the *programming literacy scale* (4 items), item difficulty ranged from .12 to .49. "I don't know" responses ranged from .21 to .42 across items. Corrected item-total correlations ranged from .11 to .29. Internal consistency was low ($\alpha = .34$), which may partly reflect the small number of items included in the scale.

C2. Subjective AI Literacy (MAILS)

The original MAILS was developed as a self-report instrument for German-speaking adults [1], and a 10-item short version was later developed to represent the main dimensions of the full scale with good model fit [3]. Because the instrument was originally developed for adults, we reviewed all items for comprehensibility and age appropriateness and used it here as an adapted self-assessment measure for a younger sample rather than a fully established child measure.

C3. Consumer-product attachment scale

The scale was originally developed to assess people's attachment to products they own (e.g., a lamp, a clock), and has also been deployed to HRI [2]. We opted to use this measure instead of other attachment measures that focus more on human-like relationship formation [e.g., 8], as we agree with recent concerns about unhealthy emotional attachment to robots [e.g., 4, 7], and believe such attachment should be treated similarly to humans' attachment to other objects rather than other humans.

C4. Student and teacher feedback

Students. We assessed students' evaluation of the full project at post-test (T7) through three researcher-developed items. Specifically, we asked students how much they enjoyed the entire project, how much they believed they learned, and how difficult they found the project on 5-point Likert scales (1 = very bad/very little/very difficult, 5 = very good/a lot/very easy). Additionally, students were invited to provide open-ended feedback on what they liked best and what they found difficult/would have liked to change.

Teachers. To assess teachers' evaluations of the two conditions and the overall project, we deployed an 8-item researcher-developed feedback questionnaire. The questionnaire was designed to capture teachers' perceptions of students' engagement and learning during each lesson, the ease of implementing the lesson, and their broader reflections on the content. Capturing teachers' perspectives on student engagement is critical, as teachers are best placed to assess it relative to overall student engagement. After each lesson, teachers responded to five Likert-scale items and three open-ended questions. The Likert-scale items assessed perceived student engagement, the number of relevant student questions, ease of guiding students through the lesson content, ease of guiding students regarding student behavior, and perceived student learning in the lesson. All items were rated on five-point scales, with higher values indicating more positive evaluations (e.g., higher engagement, more relevant questions, greater ease of implementation, and greater perceived learning). The open-ended questions asked which two aspects of the lesson teachers liked, which two aspects they found difficult or would have preferred to change, and whether they wished to provide any additional comments. These responses were collected to capture contextualized impressions of each session and to complement the quantitative ratings with more detailed process-related feedback. After both conditions concluded, teachers answered all items again, this time regarding the overall project. Due to the small number of participating teachers, we used this data to provide additional context on the deployment rather than to conduct formal statistical testing.

D. Deviations from preregistration

We deviated from our preregistration in two cases. First, to prevent overfitting and reduce the risk of false positives, we did not directly run our preregistered models (including main effects and interactions); instead, we incrementally added the preregistered effects and used likelihood ratio tests to compare models. We only continued to increase model complexity if the model fit was improved. Second, to analyze intrinsic motivation, we preregistered using a seven-level factor for individual time points. However, because our primary interest was in differences in engagement between the intervention phase (weeks 1-4) and the post-intervention phase (weeks 5-7), we instead used a simplified model with a two-level factor for phase 1/phase 2. This simplified model was conceptually better suited to address our research questions and improved the readability and interpretability of the results. The results obtained from the simplified models did not differ substantially from those of the preregistered models (available in the scripts). We additionally preregistered exploratory analyses examining the effects of intrinsic motivation (T1), attachment to the robot, and perceived anthropomorphism on children's objective learning gains. As these analyses did not yield significant results, we do not report them here; however, all analysis outputs are available on the OSF [BLINDED].

E. Descriptive analyses

Mean values and standard deviations for each group at each time point are shown in **Table S2**. Correlations among the study measures are reported in the OSF project supplements [BLINDED]. Two participants failed both attention checks (T8), and nine participants failed one attention check (T1 = 2; T7 = 3; T8 = 4). As we implemented the attention checks only after T4 in the experimental group and did not preregister to exclude participants who failed them, these participants were still included in the analyses.

Table S2. Means and Standard deviations of study measures.

	T1 <i>M (SD)</i>	T4 <i>M (SD)</i>	T7 <i>M (SD)</i>	T8 <i>M (SD)</i>
Subjective AI literacy				
Interaction	5.7 (1.00)	6.17 (1.31)	6.66 (1.45)	
Building	5.86 (2.00)	5.58 (1.40)	7.01 (1.22)	
Objective AI literacy				
Interaction	5.79 (3.86)	6.82 (4.93)	8.65 (5.04)	8.86 (4.63)
Building	5.43 (3.17)	6.48 (3.61)	7.30 (3.37)	11.00 (3.52)
	Phase 1 M (SD)		Phase 2 M (SD)	

Intrinsic motivation		
Interaction	3.77 (0.65)	3.70 (0.57)
Building	4.08 (0.68)	3.79 (0.81)

Table S3. Correlations between learning study measures.

	Objective AI literacy	Subjective AI literacy	Engagement
Objective AI literacy	1	.19	.05
Subjective AI literacy	.19	1	.05
Engagement	.05	.05	1

F. Objective AI literacy subscale analyses

For the *Technical Knowledge* subscale (**Figure S1A**), the same model comparison procedure was applied. Again, including the experimental group did not significantly improve model fit ($\chi^2(1) = 1.47$, $p = .22$), whereas adding timepoint did significantly improve model fit ($\chi^2(2) = 18.22$, $p < .001$). The interaction between timepoint and experimental group did not improve model fit compared to the main effects only model ($\chi^2(2) = 0.19$, $p = .91$) and was therefore not retained. The final model showed significant increases in *Technical Knowledge* over time. Compared to timepoint 1, scores were significantly higher at timepoint 4 ($\beta = 0.54$, $SE = 0.18$, $p = .004$) and timepoint 7 ($\beta = 0.78$, $SE = 0.18$, $p < .001$). However, post-hoc comparisons did not show a significant increase from timepoint 4 to timepoint 7 ($\beta = -0.23$, $SE = 0.19$, $p = .22$).

Model comparisons for the *Programming* subscale (**Figure S1B**), showed again a significant improve in model fit compared to the intercept only model when adding timepoint ($\chi^2(2) = 17.80$, $p < .001$) but not for experimental group ($\chi^2(1) = 0.69$, $p = .41$). The interaction between timepoint and experimental group did not significantly improve model fit compared to the main effects only model ($\chi^2(2) = 2.34$, $p = .31$) and was therefore excluded. The model including timepoint only also revealed a significant increase in *Programming* from timepoint 1 to timepoint 7 ($\beta = 0.68$, $SE = 0.16$, $p < .001$) but not from timepoint 1 to timepoint 4 ($\beta = 0.23$, $SE = 0.16$, $p = .15$). Post-hoc comparisons indicated a significant increase from timepoint 4 to timepoint 7 ($\beta = 0.45$, $SE = 0.16$, $p = .007$).

For the *AI Knowledge* subscale (**Figure S1C**), adding experimental group the intercept only model did not significantly improve model fit ($\chi^2(1) = 2.00$, $p = .16$). Adding timepoint resulted in a marginal improvement in model fit ($\chi^2(2) = 5.97$, $p = .051$). The interaction between timepoint and experimental group was not significant ($\chi^2(2) = 0.24$, $p = .89$). The final model including timepoint as a fixed effect only showed that *AI Knowledge* scores were significantly higher at timepoint 7 compared to timepoint 1 ($\beta = 0.77$, $SE = 0.32$, $p = .017$), whereas the increase from timepoint 1 to timepoint 4 was not significant ($\beta = 0.26$, $SE = 0.32$, $p = .41$). The difference between Timepoint 4 and Timepoint 7 was also not significant ($\beta = -0.51$, $SE = 0.33$, $p = .13$).

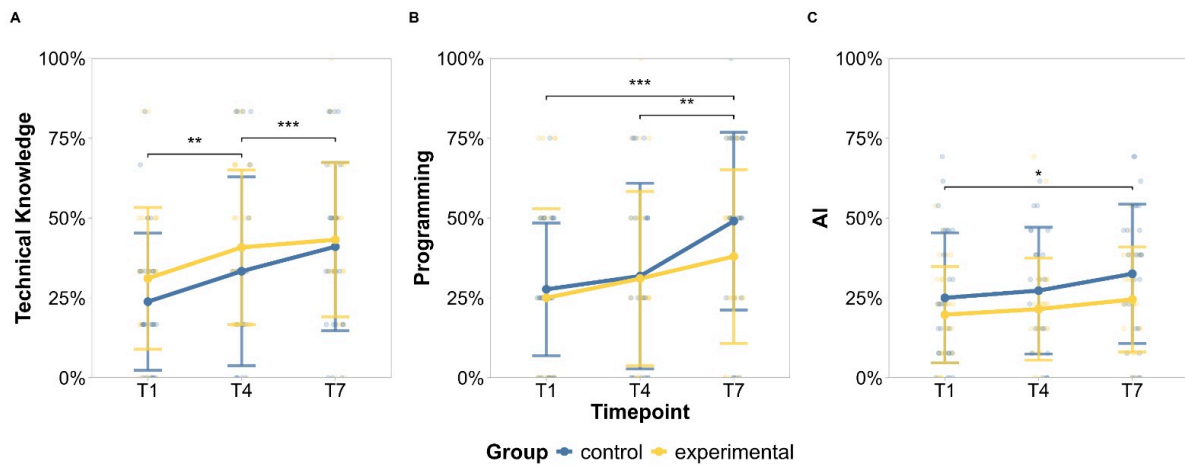


Figure S1. Changes in students' technical knowledge (A), programming knowledge (B), and AI literacy (C), over time and by group, including raw data points, means in percent, and standard deviations.

G. Perception Results (RQ3-4)

Competence (H4). Likelihood ratio tests showed no evidence that condition ($\chi^2 = 1.53, p = .22$), timepoint ($\chi^2 = 0.37, p = .83$), or their interaction ($\chi^2 = 0.39, p = .82$) predicted perceived competence (**Figure S2**). Preregistered exploratory analyses indicated that students' prior AI literacy knowledge was not associated with perceived competence ($\chi^2 = 0.11, p = .75$). However, perceived anthropomorphism significantly predicted variance in competence ($\chi^2 = 23.56, p < .001$), indicating that students who perceived the robot as more human-like regarded it to be more competent ($\beta = 0.19, SE = 0.04, p < .001$), irrespectively of condition or timepoint (**Figure S3**).

Warmth (H5). Similar to competence, likelihood ratio tests indicated students' perceived warmth was not predicted by condition ($\chi^2 = 2.78, p = .10$), timepoint ($\chi^2 = 1.11, p = .58$), or their interaction ($\chi^2 = 1.60, p = .45$; **Figure S2**). Preregistered exploratory analyses indicated that prior AI knowledge did not explain variance in students' warmth perception ($\chi^2 = 0, p = .98$). Anthropomorphism positively predicted perceived warmth ($\chi^2 = 5.02, p = .03$), with students perceiving the robot as warmer when they attributed more human-like qualities to it ($\beta = 0.08, SE = 0.04, p = .02$), independent of time and group (**Figure S3**).

Safety (H6). In lieu of previous perception measures, perceived safety was not affected by condition ($\chi^2 = 0.71, p = .38$), timepoint ($\chi^2 = 2.06, p = .36$), or their interaction ($\chi^2 = 1.14, p = .56$; **Figure S2**). Preregistered exploratory analyses indicated that students' prior AI knowledge did not affect the perceived safety of the robot ($\chi^2 = 0.17, p = .68$). However, adding anthropomorphism and timepoint significantly improved model fit ($\chi^2 = 7.91, p = .02$), with both variables positively predicting perceived safety (**Figure S3**). Pairwise contrasts showed an increase in perceived safety between T1 and T4 ($\beta = 0.66, SE = 0.26, p = .02$), but not between T4 and T7 ($\beta = -0.08, SE = 0.25, p = .76$; **Figure S3C**).

Discomfort (H7). Likelihood ratio tests showed no main effect of condition ($\chi^2 = 0.03, p = .87$) or time point ($\chi^2 = 3.56, p = .17$) on perceived discomfort. However, the interaction between condition and time point significantly improved the model ($\chi^2 = 7.35, p = .03$). Post-hoc contrasts calculated from our preregistered model indicated a significant difference in trajectories of discomfort in the control and build groups between T1 and T4 ($\beta = -1.14, SE = 0.43, p = .049$). Specifically, discomfort significantly decreased between T1 to T4 in the control group ($\beta = -0.79, SE = 0.33, p = .03$), but not in the build group ($\beta = 0.35, SE = 0.29, p = .23$; **Figure S2D**). However, because both groups did not differ significantly in perceived discomfort at T1 or T4, this interaction should be interpreted with caution. Preregistered exploratory analyses indicated that prior AI knowledge did not explain variance in students' discomfort perception ($\chi^2 = 0.01, p = .93$). However, perceived anthropomorphism

significantly improved model fit relative to the baseline model ($\chi^2 = 5.66, p = .02$). Higher perceived anthropomorphism was associated with increased perceived discomfort ($\beta = 0.08, SE = 0.03, p = .02$), irrespective of group or time (**Figure S3**).

Anthropomorphism (H8). Distinct from our other perception measures, we found evidence for a significant effect of time on perceived anthropomorphism ($\chi^2 = 19.33, p < .001$), but no effect of condition ($\chi^2 = 0.05, p = .83$), or their interaction ($\chi^2 = 1.55, p = .46$; **Figure S2**). Post-hoc contrasts indicated a decrease in anthropomorphism between T1 and T4 ($\beta = -1.68, SE = 0.43, p < .001$), but not between T4 and T7 ($\beta = 0.04, SE = 0.45, p = .93$; **Figure S3**). Preregistered exploratory analyses indicated that prior AI knowledge did not explain variance in perceived anthropomorphism ($\chi^2 = 1.08, p = .30$).

Attachment (H9). Students' attachment to the robot was not significantly affected by condition ($\chi^2 = 1.21, p = .27$), time ($\chi^2 = 4.31, p = .12$), or their interaction ($\chi^2 = 0.82, p = .66$; **Figure S2**). Exploratory analyses indicated that students' prior AI knowledge did not significantly predict their attachment to the robot ($\chi^2 < 0.001, p = .98$). Perceived anthropomorphism significantly improved model fit relative to the baseline model ($\chi^2 = 11.53, p < .001$), with higher perceived anthropomorphism being associated with greater attachment ($\beta = 0.07, SE = 0.02, p < .001$), irrespective of group or time (**Figure S3**).

Intention to use (H10). Students' intention to use was not significantly predicted by condition ($\chi^2 = 3.48, p = .06$), timepoint ($\chi^2 = 5.70, p = .06$), or their interaction ($\chi^2 = 2.91, p = .23$; **Figure S2**). Preregistered exploratory analyses indicated that attachment and condition significantly improved model fit ($\chi^2 = 4.46, p = .04$), with attachment positively predicting students' intention to use the robot ($\beta = 0.60, SE = 0.07, p < .001$), irrespective of condition or timepoint (**Figure S3**). Students in the building group showed a higher intention to use the robot than students in the control group ($\beta = 0.44, SE = 0.19, p = .02$; **Figure S3**). However, since this main effect also includes students' pre-test scores, it should be interpreted with caution. Exploratory analyses indicated that anthropomorphism significantly predicted intention to use ($\chi^2 = 14.23, p < .001$), with higher ratings associated with greater intention to use the robot, irrespective of condition or time point (**Figure S3**).

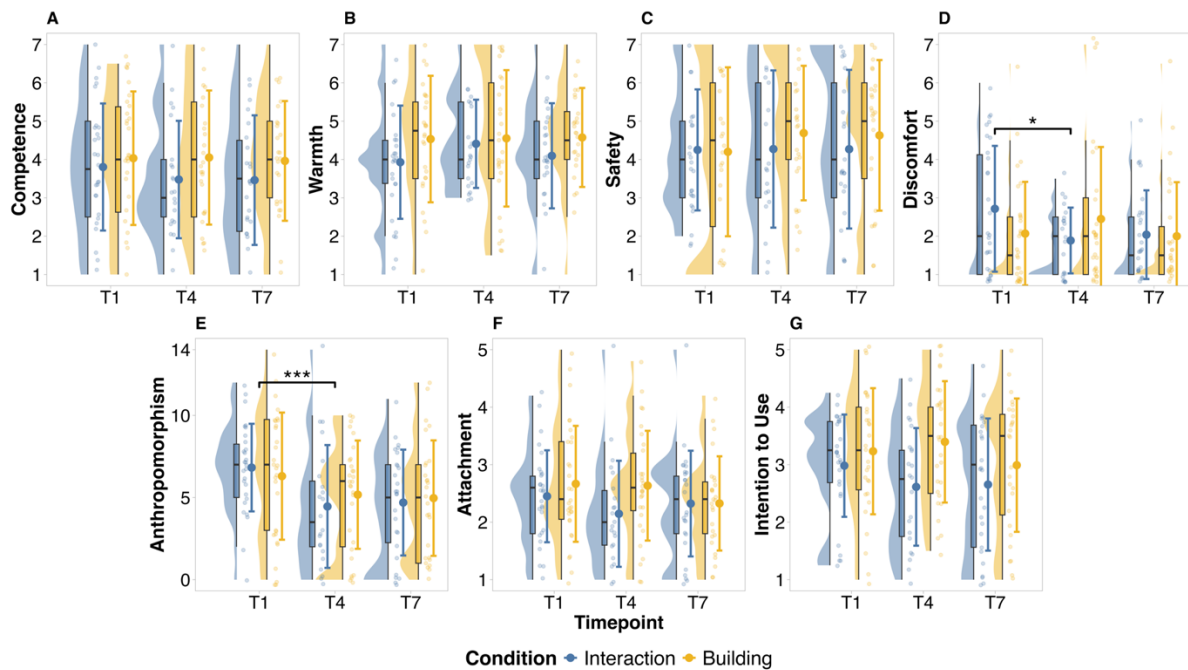


Figure S2. Means, standard deviations, and raw data points of study variables by group and condition. A) Non-significant effects on perceived competence. B) Non-significant effects on perceived warmth. C) Non-significant effects on perceived safety. D) Significant interaction effect of time and condition

on perceived discomfort. E) Significant decrease of anthropomorphism across time. F) Non-significant effects on perceived attachment. G) Non-significant effects on intention to use.

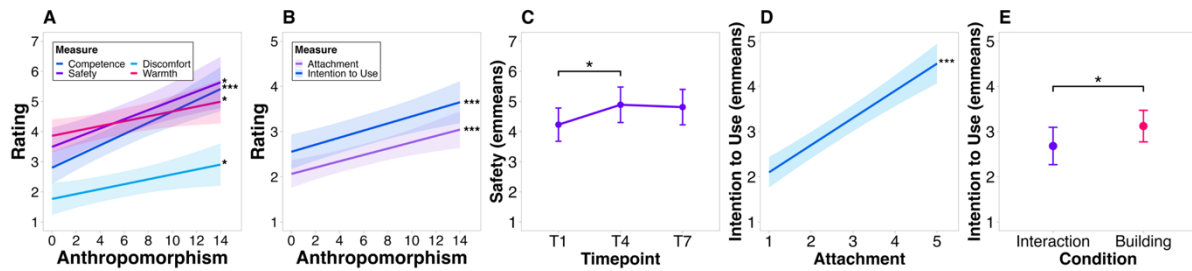


Figure S3. Model-based estimated marginal means of study variables. A-B) Significant increase in outcome variables (y-axes) with increasing anthropomorphism. C) Main effect of timepoint of safety, when controlling for anthropomorphism. D) Significant increase in intention to use with increasing attachment. E) Main effect of condition on intention to use, when controlling for attachment.

H. Students' and teachers' project evaluations

Data Analysis

In total, 882 feedback segments were coded into six categories, derived from the data, which were deemed relevant to comparing participants' experiences in the building and interaction conditions. Specifically, these categories were: (i) robot building and assembly, (ii) programming and technical exploration, (iii) Marty interaction and activities, (iv) AI-related learning and communication, (v) learner/social engagement, and (vi) technical implementation and reliability. We used the same categories for participants' positive and negative feedback to summarize both what students and teachers mostly commented on and how positively or negatively these aspects were perceived. Responses not relevant to assessing different experiences between conditions, such as no-answer responses ($n = 113$), general positive feedback (e.g., liking everything; $n = 24$), general negative feedback (e.g., not finding anything difficult; $n = 145$), general dislike responses ($n = 3$), questionnaire-related comments ($n = 12$), and instructional-design comments ($n = 53$) were excluded from the analysis and are reported elsewhere [BLINDED]. The remaining data included 532 coded segments (students: $n = 456$; teachers: $n = 76$). Student feedback included 360 positive mentions (building: $n = 177$; interaction: $n = 183$) and 96 negative mentions (building: $n = 51$; interaction: $n = 45$). Teachers' feedback included 44 positive mentions (building: $n = 24$; interaction: $n = 20$) and 32 negative mentions (building: $n = 18$; interaction: $n = 14$).

We acknowledge that content analysis is a subjective process and, as such, is affected by the analysts' experiences and perspectives. To introduce multiple perspectives, we adopted a collaborative approach in which the first author conducted the primary analysis in tandem with a student. Our interpretations were shaped by our team's diverse backgrounds in psychology, business engineering, neuroscience, cognitive sciences, and informatics, bringing together 3 to 20 years of research experience.

Results

Students' feedback

Quantitative. Children's ratings of the overall project regarding enjoyment, difficulty, and learning showed no significant differences between groups (p 's $> .16$; **Figure S4**). Relative to the 5-point scale, both groups rated the project as rather enjoyable ($M = 4.02$, $SD = 0.88$), moderately to rather difficult ($M = 3.48$, $SD = 0.97$), and reported learning neither a lot nor little ($M = 2.91$, $SD = 1.14$). These results are largely consistent with other measures in this study, which also showed no differences between groups regarding their learning or perceived pressure from the program. However, the non-significant enjoyment result does not align with the higher enjoyment/interest observed in the intrinsic motivation

subscale. On that subscale, children in the building group reported higher enjoyment/interest on average throughout the project. We discuss possible explanations for this divergence in the discussion.

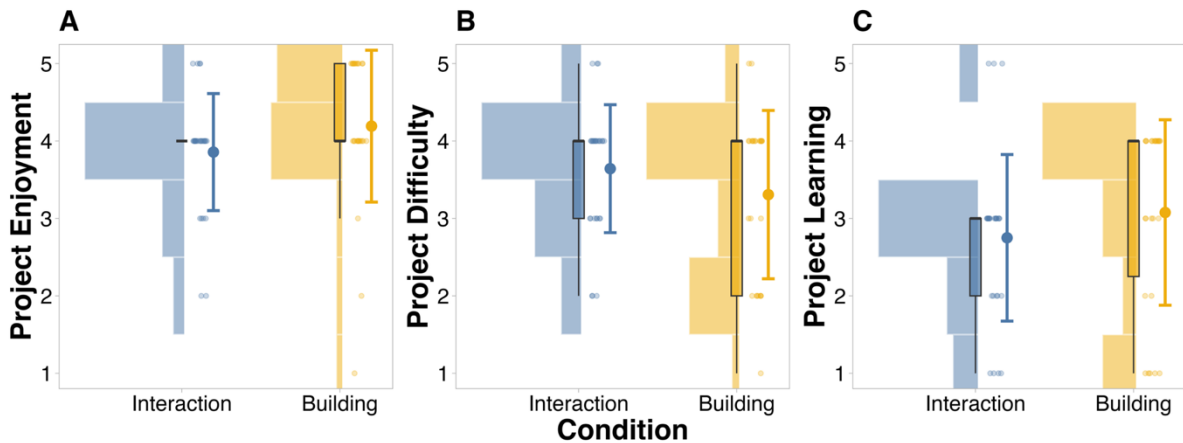


Figure S4. Students' ratings of the projects' A) enjoyment, B) difficulty, and C) perceived learning. Each plot includes bar distributions, boxplots with median, raw data points, means, and standard deviations.

Qualitative Feedback. Heat maps of students' feedback are available in **Figure S5 and S6**. During Phase 1, the two conditions involved different activities: students in the build condition built Marty, whereas students in the interaction condition interacted with pre-built Martys through activities and games. Consequently, students' positive feedback reflected these different activity structures. Students in the build condition most often highlighted robot building and assembly ($n = 74$), including building Marty ($n = 49$), robot completion ($n = 18$), setup or unpacking ($n = 2$), and progress or working speed during the build ($n = 5$). This suggests that the build condition made construction, completion, and technical progress salient as part of students' experience. In contrast, positive feedback in the interaction condition during Phase 1 primarily focused on the immediate use of Marty for interaction and activities ($n = 94$). This included activities such as the penalty shootout ($n = 19$), the obstacle course ($n = 15$), Marty's dance ($n = 9$), and the escape game ($n = 10$), as well as more general feedback around physical interaction ($n = 17$), Marty's control ($n = 8$), applying stickers or personalization ($n = 10$), and Marty's general appeal ($n = 6$).

A more informative pattern in Phase 1 was that children in the build condition mentioned programming and technical exploration ($n = 10$), whereas this category did not appear in the interaction condition. These comments mostly referred to testing or trying out Marty ($n = 7$) and programming Marty ($n = 3$), suggesting that some students may have started to experience Marty as a technical system rather than only as an interaction object. At the same time, AI-related learning and communication were mentioned more often in the interaction condition during Phase 1 (interaction: $n = 6$; build: $n = 1$), with students mainly commenting on learning or hearing about AI. This indicates that early positive feedback was not only shaped by the different activities but also by which content became salient in each condition.

As Phase 2 was designed with the same activities across conditions, differences in students' feedback during that phase may be especially informative. As expected, programming and technical exploration became equally prominent in both conditions (interaction: $n = 38$; build: $n = 38$). However, students in the build condition mentioned AI-related learning and communication more often than students in the interaction condition (build: $n = 28$; interaction: $n = 20$), whereas students in the interaction condition continued to refer more often to Marty interaction and activities (interaction: $n = 17$; build: $n = 1$). This pattern may suggest that Marty took on different roles across conditions. However, it should be interpreted cautiously, as the later scaffolded Marty/LLM activity worked as intended only in the build condition, whereas students in the interaction condition could not use the same Marty-based setup and instead engaged more generally with AI/LLM tools (see Limitations).

Learner and social engagement were mentioned slightly more often in the build condition across both phases (build: $n = 11$; interaction: $n = 8$). Such comments referred to the collaboration and

working together in a group, as well as peer-related learning, such as receiving help from peers, helping others, or showing Marty's functions to peers. This suggests that the build condition may have supported not only technical engagement, but also moments of collaborative learning and possibly done so slightly more than the interaction condition.

Negative feedback was evenly distributed across categories and between conditions, suggesting that negative comments reflected specific implementation challenges rather than a general dislike of either condition. In the build condition, Phase 1 comments mainly reflected challenges with the build process, including slow progress and calibration difficulties ($n = 2$), unclear build guides ($n = 4$), and building difficulties ($n = 5$). However, the most frequent negative building-related category was students' wish for more time to build ($n = 6$). Thus, negative feedback about building was ambivalent: some children experienced difficulties, but others appeared frustrated because they wanted to continue building rather than because they disliked the activity. This suggests that the build condition was engaging, but required sufficient time, scaffolding, and technical support.

Learner and social engagement-related negative feedback appeared in both conditions, but differed across phases. In the interaction condition, most such negative comments appeared in Phase 1, while in the building group, more were made in Phase 2. The interaction commented mainly on peer behavior during the obstacle course activity ($n = 7$), including one student's complaint about perceived unfairness. In Phase 2, negative comments about group arrangements became more prominent ($n = 4$), and one student noted that not everybody was busy during the task. This may indicate group-level difficulties, but may also reflect that some later activities did not keep all students equally involved. In the build condition, fewer children reported negative comments about group-related difficulties ($n = 2$) during Phase 1, possibly indicating that the collaborative nature of the building activity helped mitigate conflict. During Phase 2, students in the building group similarly disliked group arrangements ($n = 5$). These comments, therefore, seem to reflect classroom and group-level implementation challenges that may be more pronounced in robot interaction than in robot building.

AI-related negative feedback differed between conditions and phases. In the interaction condition, Phase 1 comments mainly concerned difficulties with the introduction to AI and with understanding the concepts ($n = 4$), whereas in the building group, no negative comments were made on this topic. This may indicate that the robot-building activity better scaffolded the AI-related content. In Phase 2, negative AI-related feedback was mostly linked to the Marty/LLM interaction not working as intended ($n = 5$), while one child found writing to the LLM boring. This fits the later implementation issues in the interaction condition, where the Marty interface did not work reliably, and students partly used alternative LLM tools instead. In contrast, in the build condition, where the scaffolded Marty interaction worked, negative AI-related comments were less about access and more about the interaction design itself ($n = 3$).

Programming-related negative feedback was rare in Phase 1 but prevalent in Phase 2, with more mentions in the build condition. The comments reflected similar concerns across conditions: some students disliked programming, while others struggled to understand or implement the programming tasks. Lastly, technical implementation and reliability issues were present in both conditions across phases, although slightly fewer comments were made in the control group during Phase 1 (interaction: $n = 3$; building: $n = 6$). Overall, the feedback mainly included robot-related issues ($n = 14$) and other problems, such as programming or interface issues ($n = 7$). This illustrates how important reliable robots and tasks are for sustaining engagement, as issues may lead to frustration and risk the success of the

deployment.

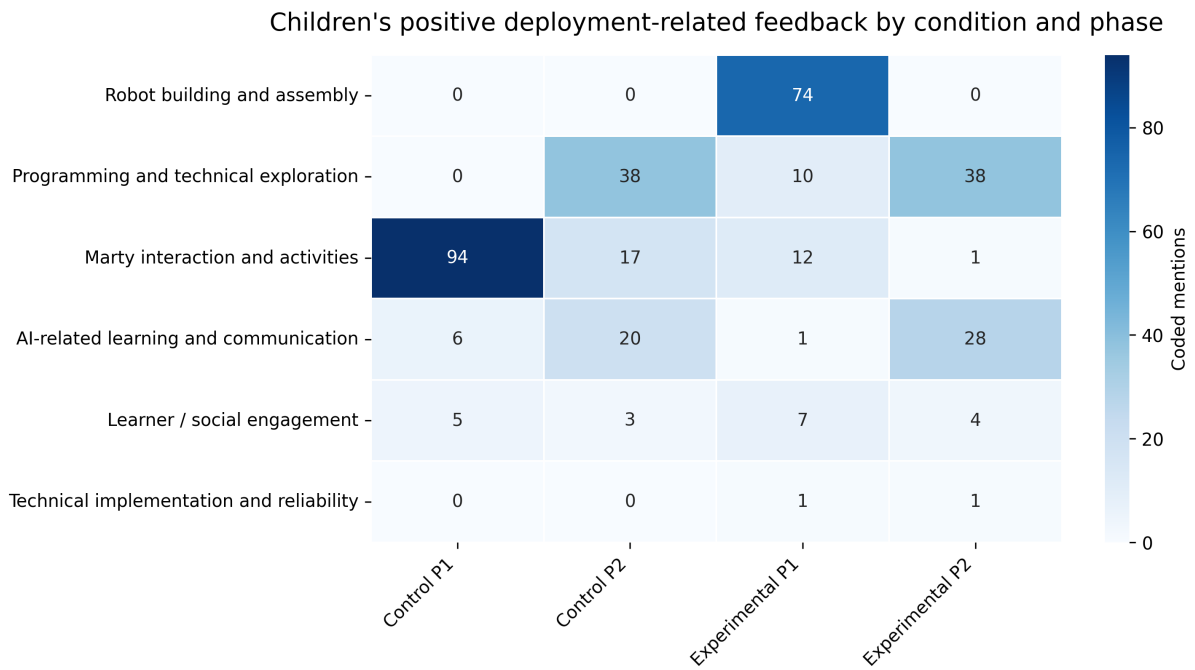


Figure S5. Heat map of coded segments of children's positive feedback by condition and phase.

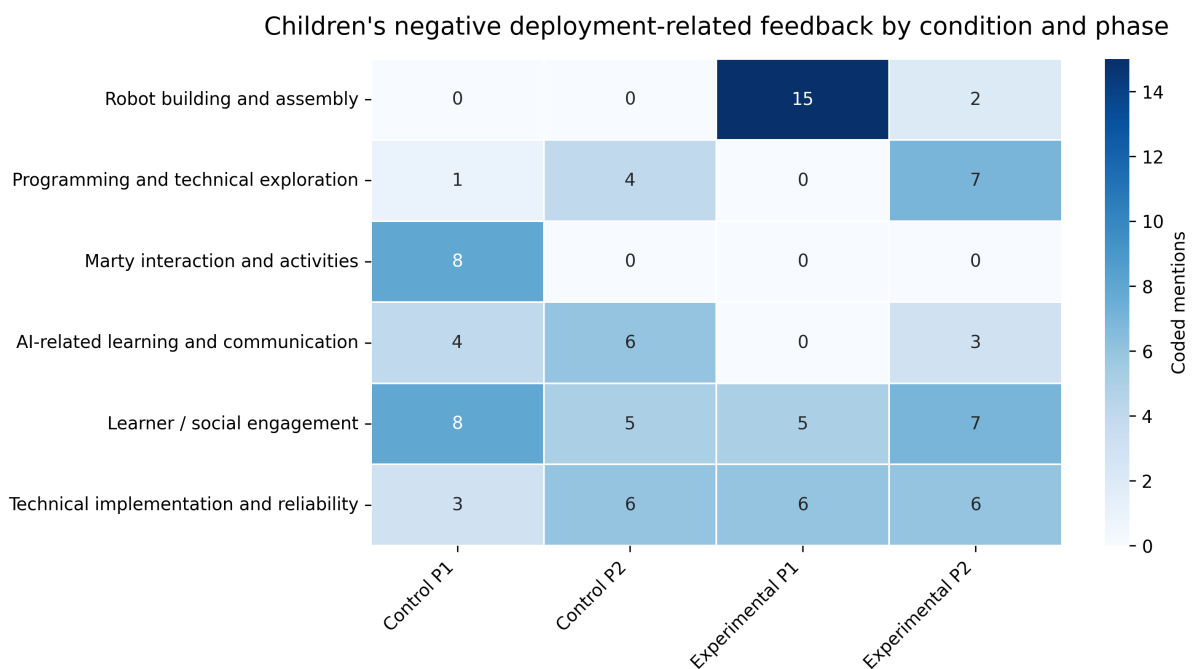


Figure S6. Heat map of coded segments of children's negative feedback by condition and phase.

Teachers' evaluations

Quantitative Feedback. Teachers' ratings of students' engagement, learning, question-asking, and the difficulty of guiding them through the lessons are shown in **Figure S7**. In the following, we discuss visual patterns in the data that may indicate differences in experiences between groups from teachers' perspectives. However, these data are descriptive and do not indicate formal differences. Means and standard deviations are available in the **Supplements** in our OSF project [BLINDED].

Teachers rated students' engagement similarly high during Phase 1, but these ratings dropped in Phase 2 for the interaction condition (**Figure S7A**). A similar drop in Phase 2 was observed regarding

relevant student questions (Figure S7B). The build condition was rated as more difficult to guide in Phase 1, both in terms of content (Figure S7C) and student behavior (Figure S7D). In Phase 2, both difficulty ratings were more similar across conditions. Students' learning was rated as moderately to highly supportive, and these ratings were relatively similar between conditions and Phases (Figure S7A).

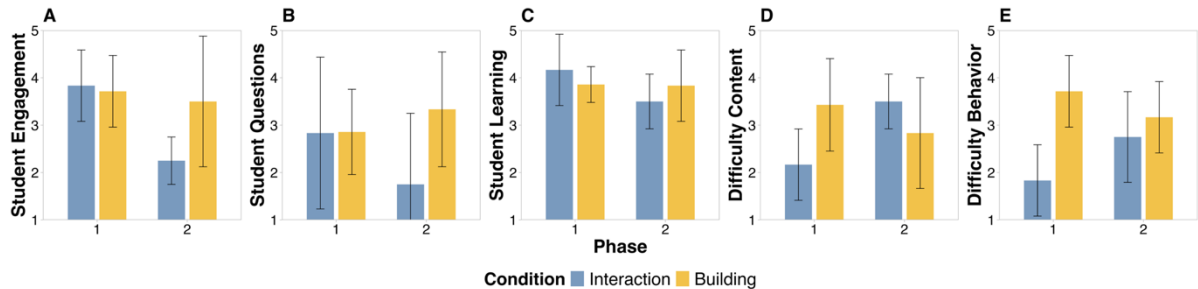


Figure S7. Descriptive teacher ratings of (A) student engagement, (B) relevant student questions, (C) perceived student learning, (D) difficulty guiding through content, (E) difficulty guiding student behavior, by phase and group. Each plot shows mean ratings averaged within phase and group, with standard deviations.

Qualitative Feedback. Heatmaps showing the occurrence of categories per condition and phase are available in Figure S8 and S9. Positive teacher feedback in the build condition primarily referred to learner and social engagement ($n = 8$). This included group work ($n = 2$), with teachers noting that students worked well together and took turns, as well as students' motivation and interest ($n = 3$). Teachers also highlighted students' focused work ($n = 2$), both during the task itself and during the theoretical part, and one comment emphasized students' opportunities to explore Marty's features independently ($n = 1$). In contrast, robot building and assembly were mentioned less frequently ($n = 2$), suggesting that teachers perceived the value of building mainly through students' engagement rather than through the act of building itself.

In the interaction condition, positive teacher feedback during Phase 1 focused on Marty interaction and activities ($n = 5$), as expected, given that students interacted with pre-built Martys through activity formats. Teachers praised the activities, including the carefully prepared obstacle course, the Marty penalty shootout, which gave students more time to control Marty directly, and the escape game, which was particularly engaging. Across the project, positive feedback concerned programming and technical exploration ($n = 7$), with teachers highlighting opportunities for students to experiment ($n = 3$), program ($n = 3$), and work practically with Marty ($n = 1$).

Negative teacher feedback in the build condition primarily concerned robot building and assembly ($n = 8$). Teachers commented on students having difficulties building the robot ($n = 2$), uneven progress between groups ($n = 2$), and difficulties teaching or supporting the activity due to students' inexperience ($n = 4$). In two cases, this was explicitly connected to teachers feeling helpless or being overwhelmed. One teacher described being "*quite overwhelmed*" because there were "*so many new components to teach*" and they could not provide "*the usual level of support when difficulties arose*", illustrating the frustration that could arise when requiring teachers to support students without sufficient technical knowledge of the robot. The teacher further worried that accumulated difficulties for students could lead to "*too much frustration*" and even "*an aversion to the subject of programming*". This suggests that successful implementation requires not only student scaffolding but also sufficient teacher preparation to respond to arising difficulties. Additional negative comments referred to technical problems ($n = 2$), problems with the randomly assigned groups ($n = 1$), and distracted students ($n = 1$).

In the interaction condition, negative teacher feedback mostly concerned learner and social engagement ($n = 5$), especially difficulties with group work and student motivation. Teachers noted that collaboration was difficult in some groups because some students withdrew from the activity. Technical implementation and reliability were also mentioned ($n = 4$), mostly in relation to Martys not working as intended. These comments highlight the importance of teacher training, classroom dynamics, and reliable technical execution for long-term school deployments.

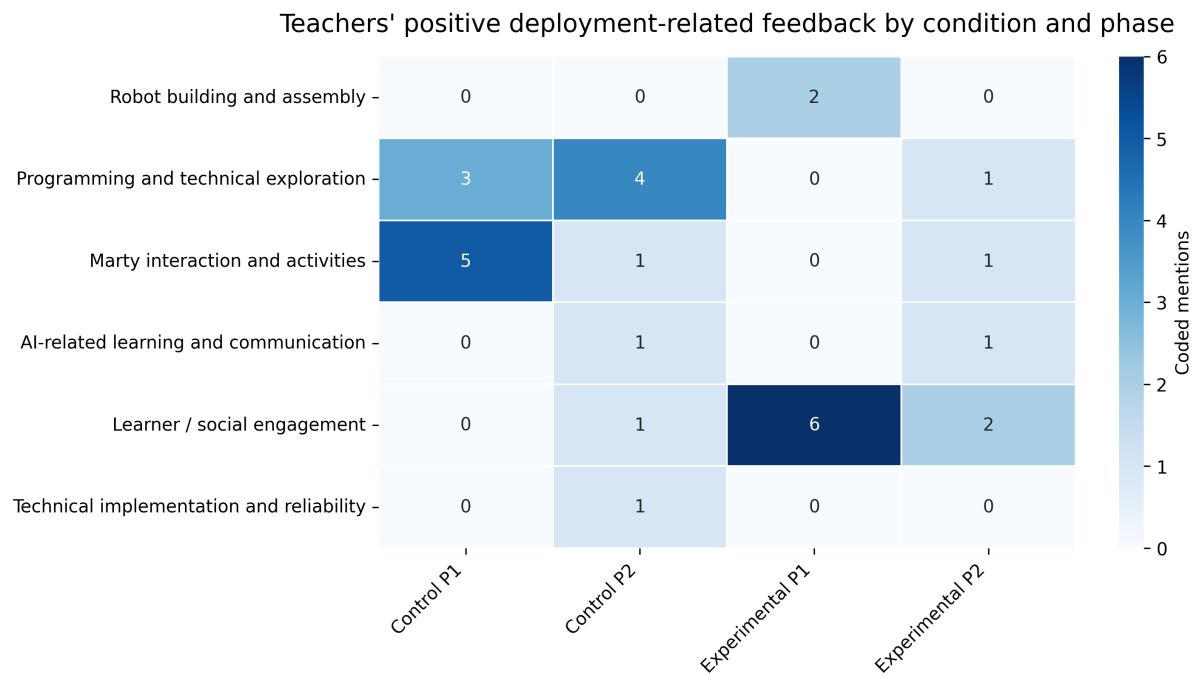


Figure S8. Heat map of coded segments of teachers' positive feedback by condition and phase.

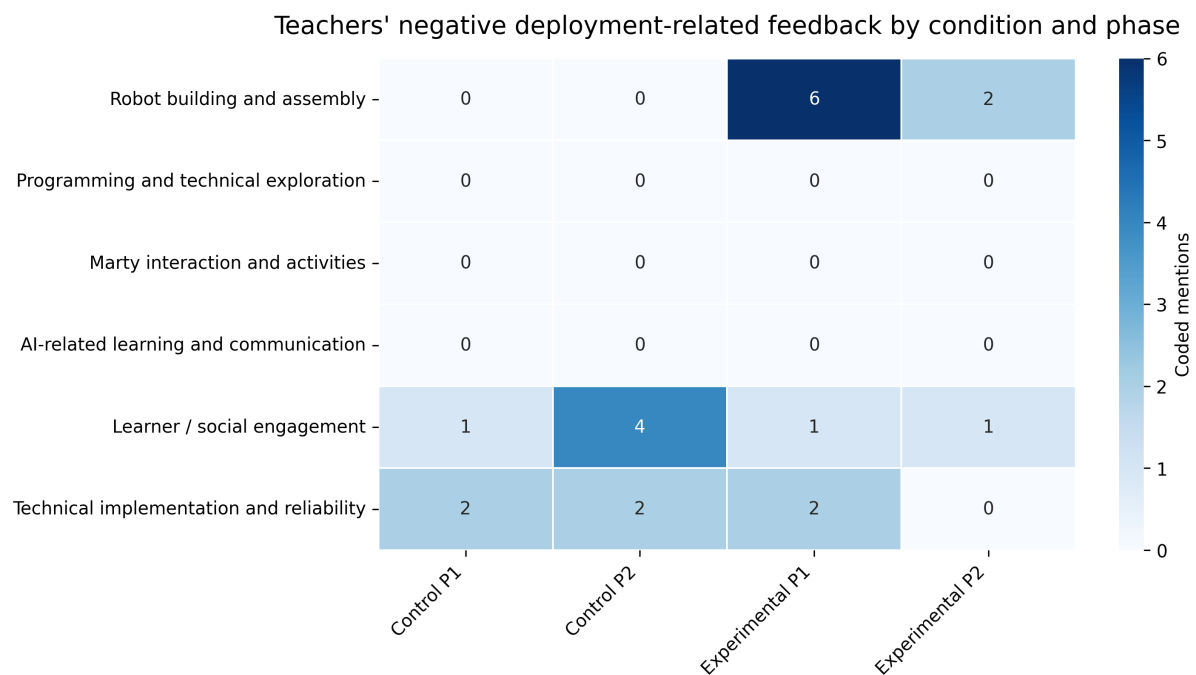


Figure S9. Heat map of coded segments of teachers' negative feedback by condition and phase.

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